Geog 322 - Lab 1 Assignment

Discussion:

Provide a comprehensive comparison and algorithm performance evaluation, including a discussion comparing/contrasting the two methods/two sensors.

Q1. Fully evaluate the two algorithms: what are their strengths? Weaknesses? What about the two different sensors? You are discussing the usefulness of the two sensors, and the usefulness of the two specific methods used in this lab for producing the SCA output. How do they differ?

Snow cover analysis can be conducted through various methods, such as the Normalized Difference Snow Index and the wet snow mask method. These algorithms have strengths and weaknesses that make them suitable for SCA analysis for different sensors. The SCA maps (*Figures 1-4*) were produced with MODIS and AMSR-E sensor data. While they are both passive sensors, they differ greatly. MODIS is a passive optical sensor that measures solar radiation (Department of Geography, University of Victoria, 2025). Optical sensors require clear weather conditions and daylight to obtain ideal imaging of Earth's surface. The temporal resolution of MODIS is 24 to 48 hours, depending on the latitude of land imaged (NASA, n.d.). Sensors and satellites that rely on solar radiation have the limitation of solely being able to read information about the surface of the land, rather than snow-pack layers or ground beneath forests. The AMSR-E sensor receives reflected microwave energy from Earth, allowing it to acquire data during day, night, and cloudy or adverse weather conditions (Department of Geography, University of Victoria, 2025). According to NASA EARTHDATA (n.d.), AMSR-E's temporal resolution is 24 to 48 hours. The Normalized Difference Snow Index from the MODIS data used green and shortwave infrared bands to distinguish snow and snow-actsfree areas, through a threshold of 0.4 (Department of Geography, University of Victoria, 2025). The NDSI uses the high reflectance of snow in green wavelengths and its high absorption in the shortwave infrared, allowing snow to be discerned from other surface types while reducing atmospheric effects through normalization (Chu, 2018). The NDSI used in this study is suitable for non-forested regions, which could lead to the misclassification of snowy areas in the North and Western regions of the study area. Utilizing wet snow masks can allow for more accurate pixel classification, since NDSI is optimized for detecting dry snow areas. Wet snow acts differently than dry snow because it absorbs more of the 18.7 GHz frequency (Tedesco & Wang, 2006). The wet snow mask method may present issues where ice is melting or groundwater is nearly frozen. When deciding between sensors and snow cover analysis methods, each choice has its benefits and drawbacks. MODIS performs best in clear, sunny conditions, and provides data with a high spatial resolution. A drawback for solar radiation sensors is that they are limited in what information they can detect, such as what lies beneath the surface, such as ground

beneath a forest. AMSR-E can collect readings at any time of day and is less affected by cloud cover and trees, compared to MODIS, due to its microwave sensors. NDSI is a simple and effective method for snow classification but requires two different formulas: one for non-forested regions and a second for forested areas. Wet snow masking allows areas that would be misclassified as non-snow-covered by NDSI to be correctly classified as snow-covered. To maximize data acquisition benefits, performing studies with two sensors and two methods of snow cover analysis provides more insightful data for research purposes.

Q2. What are potential sources of error? How could this study be improved? For example: You could think about potential differences in how you calculated these metrics compared to the studies they were adapted from. Also consider sensor limitations. The assigned papers are a good place to start.

Sources of error in this study can arise from sensors, SCA methods, and weather conditions. Data obtained from MODIS on an overcast day requires careful atmospheric correction to ensure data is as accurate as possible. Optical sensors struggle in areas that contain forests or other types of land cover. They perform best in regions like the prairies, that are fairly homogenous and flat. In the North and Western parts of the study area, it is likely that some pixels are misclassified as non-snowy, due to tree canopy cover. NDSI has two formulas that can correct data, but knowledge of the land is needed to apply the correct formula to the images. The NDSI threshold for areas without forests is 0.4 and a lower threshold is necessary for areas with vegetation (Tedesco & Miller). A threshold of 0.4 is widely used, but can be altered in different study areas for more accurate pixel classification. AMSR-E has a coarser spatial resolution than MODIS (NASA EARTHDATA, n.d.) and can result in SCA pixel values of 0, whereas the same area observed by MODIS, which has a finer spatial resolution, may be classified as snow. The Normalized Difference Snow Index does not account for areas with wet snow, but snow masking with passive microwave data can differentiate between wet snow and non-snow-covered regions. As Tedesco & Miller (2007) state, less information can be gathered from wet snow. Dry snow allows for information on depth and volume, whereas with wet snow, that information is lost. Though passive microwave sensors are less affected by the atmosphere than optical sensors, they too experience scattering from rain and water vapour (Qiu et al., 2021). This study could be improved by segmenting the study area into regions with and without forests, to correctly apply NDSI formulas and ensure passive microwave data is being used for tree cover. Misclassification would be less likely to occur with these changes, allowing for better analysis.

Q3. ***What is your final assessment? Which did a better job and why?***

There is no right or wrong answer here – both have strengths and weaknesses. Justify your choice.

Passive microwave data, combined with wet snow masking performed better. Though MODIS data classified a lower percentage of snow-covered land in February, it is possible that the coarser spatial resolution of AMSR-E homogenized areas with very little snow into a snow pixel. When observing *Figures 1 and 4*, MODIS data for May is likely classifying melting snow as a non-snow-covered area, while AMSR-E data can detect the difference in the dry snow, likely at higher altitudes, and wet snow in lower altitudes. Passive microwave sensors are less affected by atmospheric conditions, tree cover, and time of data acquisition. Optical sensors cannot provide information on snow volume or depth and therefore, imaging from AMSR-E is far more versatile for snow analysis, despite having a coarser spatial resolution.

Part 2: Deliverables

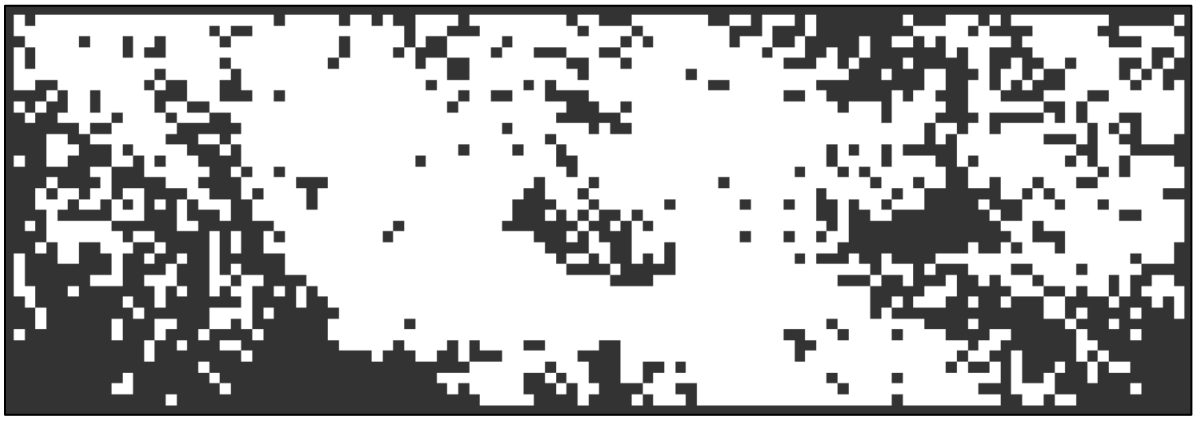

Figure 1. MODIS snow cover area classification from February 2011, with white pixels representing snow-covered land and black pixels representing areas without snow.

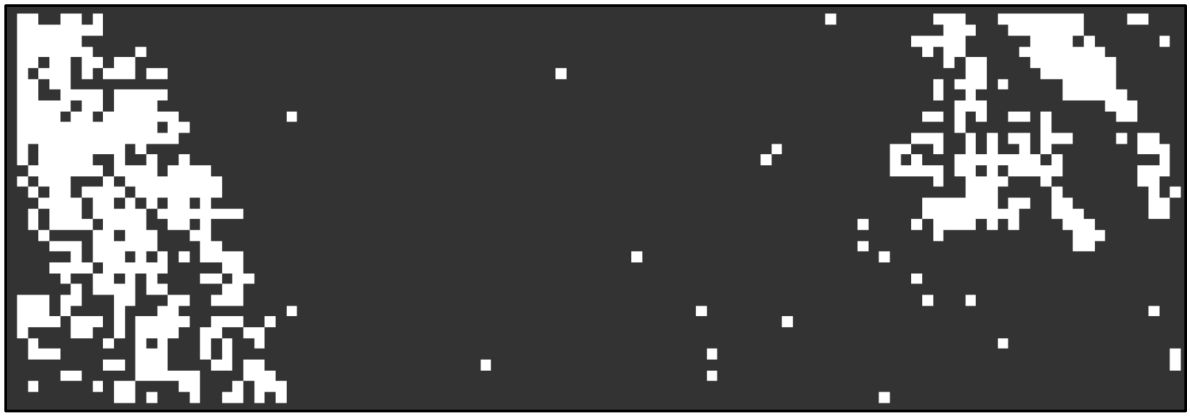

Figure 2. MODIS snow cover area classification from May 2011, with white pixels representing snow-covered land and black pixels representing areas without snow.

Figure 3. Snow cover area classification derived from February 2011 AMSRE data, with white and black pixels demonstrating snow-covered and snow-free surfaces, respectively.

Figure 4. Snow cover area classification derived from May 2011 AMSRE data, where white and black pixels demonstrating snow-covered and snow-free surfaces, respectively.

Table 1. Snow cover area information derived from AMSRE and MODIS data for February and May 2011.

	AMSRE		**MODIS**	
	February	May	February	May
snow pixels	3,800	1,231	2,529	609
snow %	97.7	31.7	65	15.7
ground area km^2	855000	276,975	569,025	137,025
clear pixels	88	2,657	1,359	3279
clear %	2.3	68.3	35	84.3
ground area km^2	19,800	597,825	305,775	737,775

References:

Chu, D. (2018). Spatiotemporal variability of snow cover on Tibet, China using MODIS remote-sensing data. *International Journal of Remote Sensing*, *39*(20), 6784–6804. https://doi.org/10.1080/01431161.2018.1466075

Department of Geography, University of Victoria. (2025). *GEOG 322 Lab 1: Passive microwave vs. optical imagery for snow detection*. https://bright.uvic.ca/content/enforced/461662-202601GEOG322A01B01-B03XCO/322-Lab1-Tutorial.pdf?ou=461662

NASA. (n.d.). *About MODIS (Moderate Resolution Imaging Spectroradiometer)*. NASA. Retrieved January 30, 2026, from https://modis.gsfc.nasa.gov/about/

NASA EARTHDATA. (n.d.). *Advanced Microwave Scanning Radiometer EOS (AMSR-E)*. NASA. https://www.earthdata.nasa.gov/data/instruments/amsr-e

Qiu, Y., Shi, L., Lemmetyinen, J., Shi, J., & Wang, R. Y. (2021). Atmospheric Correction to Passive Microwave Brightness Temperature in Snow Cover Mapping Over China. *IEEE Transactions on Geoscience and Remote Sensing*, *59*(8), 6482–6495. https://doi.org/10.1109/TGRS.2020.3031837

Tedesco, M., & Wang, J. R. (2006). Atmospheric correction of AMSR-E brightness temperatures for dry snow cover mapping. *IEEE Geoscience and Remote Sensing Letters, 3*(3), 352–356. https://doi.org/10.1109/LGRS.2006.874391

Tedesco, M., & Miller, J. (2007). Northern Hemisphere Snow-Covered Area Mapping: Optical Versus Active and Passive Microwave Data. *IEEE Geoscience and Remote Sensing Letters*, *4*(2), 221–225. https://doi.org/10.1109/LGRS.2006.888842